

Lean Six Sigma Green Belt, Glossary of Terminology

The Manufacturing Academy

5 Whys: A root cause analysis technique that involves asking "why" five times to uncover the underlying cause of a problem.

5S: A workplace organization method focusing on Sort, Set in Order, Shine, Standardize, and Sustain to improve efficiency.

Affinity Diagram: A tool used to organize ideas into groups based on their natural relationships, often used in brainstorming sessions.

Andon: A visual control system used in manufacturing to signal when there is a problem on the production floor.

ANOVA (Analysis of Variance): A statistical method used to compare means of different groups and determine if there are significant differences.

Benchmarking: The process of comparing business processes and performance metrics to industry best practices from other companies.

Black Belt: A professional trained to an advanced level in Lean Six Sigma methodologies, focusing on leading complex improvement projects.

Box Plot: A graphical representation of data that shows the distribution through their quartiles, highlighting median and outliers.

Brainstorming: A creative problem-solving technique used to generate a large number of ideas or solutions.

Business Process Mapping: A visual representation of the steps involved in a business process to improve understanding and identify inefficiencies.

Capability Index (Cpk): A statistical measure of process capability, showing how well a process can produce output within specification limits.

Cause and Effect Diagram: A tool used to identify potential causes of problems, also known as a fishbone or Ishikawa diagram.

Cellular Manufacturing: A method of arranging workstations in close proximity to reduce movement and increase efficiency.

Change Management: The structured approach to managing changes in an organization, including new processes, technologies, or behaviors.

Chi-Square Test: A statistical test used to determine if there is a significant association between two categorical variables.

Control Chart: A graph used to monitor the stability of a process over time and detect variations outside of normal limits.

Control Limits: The upper and lower bounds in a control chart that indicate the expected range of process variation.

COPQ (Cost of Poor Quality): The total cost associated with not producing a quality product or service, including rework, scrap, and customer dissatisfaction.

Correlation: A statistical relationship between two variables, where changes in one variable are associated with changes in another.

Critical to Quality (CTQ): Key characteristics of a product or process that have a significant impact on its quality and customer satisfaction.

Cycle Time: The total time it takes to complete one cycle of a process, from start to finish.

Data Collection Plan: A structured approach to gathering data that ensures consistency, accuracy, and relevance to the problem being solved.

Defect: Any instance where a product or service fails to meet the defined specifications or customer expectations.

Define Phase: The first phase of the DMAIC methodology, where the project goals and customer requirements are clearly defined.

Design of Experiments (DOE): A systematic method to determine the relationship between factors affecting a process and the output of that process.

Design Thinking: A human-centered approach to problem-solving that emphasizes empathy, ideation, and prototyping.

DFMEA (Design Failure Mode and Effects Analysis): A structured approach to identifying potential failures in a design and their effects, often used to improve reliability.

DMAIC: A data-driven, five-phase methodology used to improve processes: Define, Measure, Analyze, Improve, and Control.

DPMO (Defects per Million Opportunities): A measure of process performance, indicating how many defects occur per million opportunities.

Fishbone Diagram: Another name for a cause and effect diagram, used to identify root causes of a problem.

Flowchart: A diagram that represents the steps in a process, used to visualize workflows and identify bottlenecks.

FMEA (Failure Mode and Effects Analysis): A systematic tool for evaluating potential failure modes within a process and their impact on performance.

Gage R&R (Repeatability and Reproducibility): A statistical tool used to evaluate the precision of a measurement system.

Gemba: A Japanese term meaning "the real place," referring to the location where value is created, often used in lean to emphasize the importance of going to the worksite to observe processes.

Green Belt: A professional trained in Lean Six Sigma methodologies, typically involved in process improvement projects at a moderate level of complexity.

Hoshin Kanri: A strategic planning process used to align company goals and initiatives across all levels of an organization.

Hypothesis Testing: A statistical method used to test an assumption about a population parameter based on sample data.

Ishikawa Diagram: Another name for a cause and effect or fishbone diagram, used for identifying the root causes of problems.

Jidoka: A lean manufacturing principle that focuses on building quality into the production process by stopping it when a defect is detected.

Kaizen: A Japanese term meaning "continuous improvement," referring to small, incremental changes to processes for better efficiency.

Kanban: A visual scheduling system used in lean manufacturing to manage inventory levels and ensure a smooth production flow.

Kano Model: A framework used to categorize customer preferences into different levels of satisfaction, from basic needs to delighters.

Key Performance Indicator (KPI): A measurable value that demonstrates how effectively a company is achieving its objectives.

Lean: A methodology focused on minimizing waste while maximizing value in business processes.

Lean Six Sigma: A hybrid approach that combines lean manufacturing principles and Six Sigma techniques to improve efficiency and reduce defects.

Measure Phase: The second phase of the DMAIC methodology, focusing on measuring current process performance.

Measurement System Analysis (MSA): A statistical analysis used to assess the accuracy and reliability of a measurement system.

Monte Carlo Simulation: A mathematical technique used to model the probability of different outcomes in a process, based on random variables.

Muda: A Japanese term meaning "waste," referring to any activity that consumes resources without adding value in lean manufacturing.

NEMAWASHI: A Japanese term for the process of gaining consensus before implementing a change, often through informal discussions with stakeholders.

Normal Distribution: A bell-shaped probability distribution that is symmetrical around the mean, used in statistics for process data analysis.

Null Hypothesis: A default hypothesis in statistical testing that assumes no effect or no difference between groups.

One-Piece Flow: A lean manufacturing concept where products are produced one at a time, without waiting in batches, to reduce lead time and inventory.

Pareto Chart: A bar graph that represents the frequency of defects or causes in descending order, often used to identify the most significant issues.

PDCA (Plan-Do-Check-Act): A continuous improvement cycle that involves planning, implementing, reviewing, and acting on improvements.

Poka-Yoke: A mistake-proofing technique used in lean manufacturing to prevent defects by designing processes that make errors impossible or immediately detectable.

Process Capability: A measure of how well a process can produce products within specification limits, often expressed as a capability index (Cpk).

Process Map: A visual representation of the steps involved in a process, used to analyze and improve workflow efficiency.

Pull System: A lean manufacturing strategy where production is driven by actual demand rather than forecasted demand.

Pugh Matrix: A decision-making tool used to evaluate and compare multiple design options against a set of criteria.

RACI Matrix: A responsibility assignment chart used to clarify roles and responsibilities in a project or process.

Regression Analysis: A statistical method used to determine the relationship between a dependent variable and one or more independent variables.

RFP (Request for Proposal): A document used to solicit proposals from potential suppliers for a project or service.

Root Cause: The underlying reason for a problem or defect, typically identified through root cause analysis tools like the 5 Whys or fishbone diagram.

Run Chart: A graph that shows how a process performs over time, used to identify trends and variations in data.

Scatter Plot: A graphical representation of two variables, used to identify relationships or correlations between them.

SIPOC Diagram: A tool used to map a process by identifying Suppliers, Inputs, Process, Outputs, and Customers.

Six Sigma: A data-driven methodology aimed at improving process quality by reducing variability and defects.

SMED (Single-Minute Exchange of Dies): A lean manufacturing technique used to reduce setup times for equipment, enabling quicker changeovers.

SPC (Statistical Process Control): A method used to monitor and control processes using statistical tools, often through control charts.

Stakeholder Analysis: A process used to identify and assess the interests and influence of stakeholders in a project or process.

Standard Deviation: A statistical measure of variability or dispersion in a set of data, showing how spread out the values are.

Standard Work: A lean manufacturing concept that defines the most efficient way to perform a task, ensuring consistency and quality.

SWOT Analysis: A strategic planning tool used to identify Strengths, Weaknesses, Opportunities, and Threats in a business or project.

Takt Time: The rate at which products must be produced to meet customer demand, calculated by dividing available time by demand.

Theory of Constraints (TOC): A management philosophy that focuses on identifying and improving the most critical constraint limiting process performance.

Value Stream: The entire series of activities required to deliver a product or service to a customer, from raw materials to final delivery.

Value Stream Mapping: A lean tool used to visualize and analyze the flow of materials and information

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